Lecture 19 — Matrix-vector multiplication

Wed Oct 7

Last updated: October 5, 2026.

Boyd video (optional): Lecture 18 — matrix–vector multiplication · Lecture 19 — matrix–vector examples · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 6.4. Related objectives: Lectures. Schedule. Connects directly to Week 1 linear combinations.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, students should be able to:

1. Compute Ax by hand for a small matrix.
2. Interpret Ax as a linear combination of the columns of A.
3. Interpret A as an input-output transformation.
4. Explain how the dimensions of A determine the input and output spaces.

Warm-up

Let $a = (1, 2)$ and $b = (-1, 1)$. Compute $2a - 3b$. (This is the same skill as "columns of A times entries of x.")

Fill in

If A is $m \times n$ and x is an n-vector, the product $y = Ax$ is an _____-vector.

Row view. The ith entry of Ax is the inner product of x with the _____ of A:

$$y_i = (\text{row } i)^{\mathsf{T}} x.$$

Column view (main story). If $a_1, \dots, a_n$ are the columns of A, then

$$Ax = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n.$$

So Ax is a _____ of the columns of A, with coefficients from x.

Dimensions: A maps inputs in $\mathbb{R}$—— to outputs in $\mathbb{R}$——. Thinking of A as a machine: feed in x, get out Ax.

Special cases: $Ix = x$; $0x = 0$; if $A = \text{diag}(\lambda_1, \dots, \lambda_n)$, then $Ax = (\lambda_1 x_1, \dots, \lambda_n x_n)$.

Worked in class

W1. Compute by hand.

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

Find Ax using both the row-inner-product view and the column-combination view.

W2. Same A, different x. Without recomputing from scratch: if you already know Ax from W1, what is $A(2x)$? What is $A(x + e_1)$ if $e_1 = (1, 0, 0)$? (Use linearity / columns.)

W3. Input-output story. A 2×2 matrix

$$A = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$$

sends (urban, rural) population fractions today to tomorrow. If $x = (0.6, 0.4)$, compute Ax and interpret the two output entries.

W4. Dimension mismatch. Can you form Ax if A is 3×2 and x is a 3-vector? What size must x be?

Your turn

Y1. Compute

$$\begin{bmatrix} 2 & -1 \\ 0 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ -2 \end{bmatrix}.$$

Y2. Let A have columns $a_1 = (1, 0, 1)$ and $a_2 = (0, 1, 1)$. Write A explicitly. For $x = (3, -2)$, express Ax as a linear combination of a_1, a_2, then evaluate.

Y3. True or false: if A is $m \times n$, then Ax is always in the span of the columns of A.

Y4. A sensor turns a 3-vector of raw readings x into a 2-vector of summaries via Ax. What are the dimensions of A? In one sentence, what do the columns of A mean?

Answers (Your turn)

Y1. $(2)(4) + (-1)(-2) = 10$; $(0)(4) + (3)(-2) = -6$; $(1)(4) + (1)(-2) = 2$. So $Ax = (10, -6, 2)$.

Y2. $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$. $Ax = 3a_1 - 2a_2 = (3, -2, 1)$.

Y3. **True.** By the column interpretation, Ax is always a linear combination of the columns.

Y4. A is 2×3. The jth column is the summary 2-vector produced when the input is the standard unit vector e_j (effect of "only feature j active").